Supplementary Materials for Chapter 5.

Case and matched control comparison of comorbidities in juvenile-onset rheumatic diseases in the English National Health Service: a population-based study

Shall be considered in conjunction with the

Thesis submitted in accordance with the requirements of the

University of Liverpool

for the degree of Doctor of Philosophy

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Supplementary Table S5.1: A list of 213 comorbidities investigated in the study. 3

Supplementary Table S5.2: Medcodeid used to define Juvenile Idiopathic Arthritis. Note that where adult rheumatic codes have been included (e.g. Rheumatoid arthritis), we also specified that the individual was under 18 at the point at which the code was observed. 8

Supplementary Table *S5.*3: Medcodeid used to define Juvenile-onset Systemic Lupus Erythematosus. Note that where adult lupus codes have been included, we also specified that the individual was under 18 at the point at which the code was observed. 17

Supplementary Table S*5.*4: Medcodeid used to define Juvenile Dermatomyositis. Note that where adult codes have been included, we also specified that the individual was under 18 at the point at which the code was observed. 18

Supplementary Table S5.5: ICD-10 codes for the following comorbidities were collated from published code lists and confirmed by the study team. 19

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Supplementary Table S5.1: A list of 213 comorbidities investigated in the study.

Comorbidity
Alopecia areata
Abdominal aortic aneurysm
Autism and asperger's syndrome
Atrioventricular blocks
Angiodysplasia of colon
Acne
Allergic and chronic rhinitis
Atrial fibrillation
Abdominal hernia
Anterior and intermediate uveitis
Ankylosing spondylitis
Alcohol misuse
Alcoholic liver disease
Autonomic neuropathy
Anxiety disorders
Anaemia_other
Aplastic anaemias
Asbestosis
Asthma
Autoimmune liver disease
Bipolar affective disorder and mania
Benign essential tremor
Barrett's oesophagus
Benign prostatic hyperplasia
Bronchiectasis
Cardiomyopathy_other
Cataract
Chronic cystitis
Cystic fibrosis
Chronic fatigue syndrome
Coronary heart disease (not otherwise specified)
Cirrhosis
Chronic kidney disease
Coeliac disease
Collapsed vertebra
COPD
Cerebral Palsy
Crohn's disease
Chronic sinusitis
Congenital septal defect
Chronic viral hepatitis

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Dilated cardiomyopathy
Diverticular Disease
Dementia
Depression
Dermatitis
Diaphragmatic hernia
Diabetes mellitus_other or not specified
Diabetic eye disease
Diabetic neuropathy
Down's syndrome
Dysmenorrhoea
Enteropathic arthropathy
Eating Disorders
Endometriosis
Endometrial hyperplasia and hypertrophy
Epilepsy
Enthesopathy and synovial disorder
Folate deficiency anaemia
Fibromatosis
Fatty liver
Giant cell arteritis
Gastritis and duodenitis
Glaucoma
Glomerulonephritis
Gastro-oesophageal reflux disease
Gout
Hearing loss
Hepatic failure
Heart failure
HIV
Hodgkin lymphoma
Hidradenitis suppurativa
Hypertrophic cardiomyopathy
Hyperkinetic disorders
Hypertrophic nasal turbinates
Hyperparathyroidism
Hypersplenism
Hypertension
Hyposplenism
Idiopathic Intracranial Hypertension
Immunodeficiencies
Infections
Infertility
Intellectual disability
Intervertebral disc disorders

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Intracerebral haemorrhage
Irritable bowel syndrome
Ischaemic stroke
Left bundle branch block
Leukaemia
Low HDL-C
Lupus erythematosus
Macular degeneration
Meniere's disease
Migraine
Motor neuron disease
Multiple sclerosis
Multiple valve disorder
Myasthenia gravis
Myelodysplastic Syndrome
Myocardial Infarction
Nonrheumatic aortic valve disorders
Neuropahtic bladder
Non-Hodgkin lymphoma
Nonrheumatic mitral valve disorders
Obesity
Obsessive-compulsive disorder
Oesophageal varices
Osteoarthritis (excl spine)
Osteoporosis
Other haemolytic anaemias
Pancreatitis
Parkinson's disease
Personality disorders
Pericardial effusion
Peripheral neuropathy
Peripheral vascular disease
Plasma cell malignancy
Pleural effusion
Pleural plaque
Polycystic ovarian syndrome
Polymyalgia rheumatica
Polycythaemia vera
Posterior uveitis
Portal hypertension
Primary malignancy_Bladder
Primary malignancy_Bone
Primary malignancy_Brain
Primary malignancy_Breast
Primary malignancy_Biliary Tract

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Primary malignancy_Bowel
Primary malignancy_Cervix
Primary malignancy_Kidney
Primary malignancy_Lung
Primary malignancy_Liver
Primary malignancy_Melanoma
Primary malignancy_Mesothelioma
Primary malignancy_Multiple Sites
Primary malignancy_Oesophageal
Primary malignancy_Oropharyngeal
Primary malignancy_other
Primary malignancy_Ovary
Primary malignancy_Pancreas
Primary malignancy_Prostate
Primary malignancy_Skin
Primary malignancy_Stomach
Primary malignancy_Testis
Primary malignancy_Thyroid
Primary malignancy_Uterus
Primary pulmonary hypertension
Primary thrombocytopaenia
Psoriatic arthritis
Psoriasis
Pulmonary fibrosis
Right bundle branch block combinations
Rheumatoid arthritis
Raised LDL-C
Raynaud's syndrome
Obstructive and reflux uropathy
Rosacea
Raised triglycerides
Raised total cholesterol
Rheumatic valve disorder
Retinal vascular occlusions
Sarcoidosis
Subarachnoid haemorrhage
Substance misuse
Sickle cell disease
Schizophrenia
Scleroderma
Scleritis and episcleritis
Seborrheic dermatitis
Secondary malignancy_adrenal gland
Secondary malignancy_bone
Secondary malignancy_brain

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Secondary malignancy_Bowel
Secondary malignancy_Lung
Secondary malignancy_Liver
Secondary malignancy_Lymph Nodes
Secondary malignancy_other
Secondary malignancy_Pleura
Secondary malignancy_Peritoneum
Secondary pulmonary hypertension
Secondary polycythaemia
Secondary thrombocytopaenia
Sjogren's syndrome
Sleep apnoea
Spina bifida
Spondylosis
Spondylolisthesis
Supraventricular tachycardia
Spinal stenosis
Sick sinus syndrome
Stable angina
Stroke_not otherwise specified
Type 1 diabetes mellitus
Type 2 diabetes mellitus
Thalassaemia
Thrombophilia
Tinnitus
Transient ischaemic attack
Trigeminal neuralgia
Tubulo-interstitial nephritis
Thyroid disease
Ulcerative colitis
Unstable angina
Urinary incontinence
Visual impairment and blindness
Vitamin B12 deficiency anaemia
Vitiligo
Ventricular tachycardia
Venous thromboembolic disease (Excl PE)

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Supplementary Table S5.2**: Medcodeid used to define Juvenile Idiopathic Arthritis. Note that where adult rheumatic codes have been included (e.g. Rheumatoid arthritis), we also specified that the individual was under 18 at the point at which the code was observed.**

Medcodeid	Term	Original Read Code	Cleansed Read Code	Snomed CT Concept Id	Snomed CT Description Id
309829012	Other juvenile arthritis	N045	N045.00	239796000	2.7401E+13
312523013	[X]Other juvenile arthritis	Nyu15	Nyu1500	239796000	4.1135E+14
483981000006116	Ankylosis of multiple joints	N0859	N085900	90242006	4.8398E+14
178449010	Ankylosis of joint	N085	N085.00	111227009	178449010
310553013	Ankylosis of the subtalar joint	N085R	N085R00	202319005	310553013
310557014	Ankylosis of joint NOS	N085z	N085z00	111227009	4.8395E+14
310539013	Ankylosis of the elbow joint	N085C	N085C00	202307002	310539013
310550011	Ankylosis of other pelvic joint	N085N	N085N00	267934008	4.8401E+14
310554019	Ankylosis of other tarsal joint	N085S	N085S00	267936005	4.8402E+14
310543012	Ankylosis of the wrist joint	N085F	N085F00	202310009	310543012
483971000006119	Ankylosis of MTP joint	N085T	N085T00	202321000	4.8397E+14
310556017	Ankylosis of toe joint	N085U	N085U00	202322007	310556017
359848019	Ankylosis of spine NOS	N14z-1	N14z.11	47049007	4.8404E+14
890391000006115	Ankylosis - shoulder joint	N0851-99	N085199	202298000	8.9039E+14
890451000006119	Ankylosis - ankle/foot	N0857-99	N085799	267936005	8.9045E+14
483991000006118	Ankylosis of other CMC joint	N085H	N085H00	202310009	4.8399E+14
484071000006116	Ankylosis of the ankle joint	N085Q	N085Q00	202318002	4.8407E+14
484121000006116	Ankylosis of the shoulder joint	N085A	N085A00	202298000	4.8412E+14
890411000006115	Ankylosis - wrist joint	N0853-99	N085399	267933002	8.9041E+14

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890431000006114	Ankylosis - hip joint	N0855-99	N085599	267934008	8.9043E+14
890441000006116	Ankylosis - knee joint	N0856-99	N085699	267935009	8.9044E+14
484031000006119	Ankylosis of PIP joint	N085K	N085K00	202314000	4.8403E+14
484111000006112	Ankylosis of the knee joint	N085P	N085P00	202317007	4.8411E+14
484051000006114	Ankylosis of teeth	J016	J016.00	14901003	4.8405E+14
484141000006111	Ankylosis of the superior radio-ulnar joint	N085D	N085D00	202308007	4.8414E+14
890371000006116	Ankylosis - multiple joint	N0849-98	N084998	202264009	8.9037E+14
890401000006118	Ankylosis - elbow joint	N0852-99	N085299	267932007	8.904E+14
310538017	Ankylosis of other joint of the shoulder girdle	N085B	N085B00	302154007	4.84E+14
483961000006114	Ankylosis of MCP joint	N085J	N085J00	202313006	4.8396E+14
484171000006115	Ankylosis/instability of cervical,thoracic or lumbar spine	N148	N148.00	47049007	4.8417E+14
890461000006117	Ankylosis - other joint	N0858-99	N085899	7.0154E+14	8.9046E+14
890421000006111	Ankylosis - hand joint	N0854-99	N085499	56923001	8.9042E+14
483931000006117	Ankylosis of DIP joint	N085L	N085L00	202315004	4.8393E+14
484061000006111	Ankylosis of the 1st CMC joint	N085G	N085G00	202311008	4.8406E+14
484091000006115	Ankylosis of the hip joint	N085M	N085M00	24273009	4.8409E+14
484101000006114	Ankylosis of the inferior radio-ulnar joint	N085E	N085E00	202309004	4.841E+14
359321011	Distal interphalangeal psoriatic arthropathy	M1601	M160100	239812005	359321011
312521010	[X]Other psoriatic arthropathies	Nyu13	Nyu1300	33339001	4.1308E+14
312518013	[X]Other seropositive rheumatoid arthritis	Nyu11	Nyu1100	239791005	4.1353E+14
312520011	[X]Other specified rheumatoid arthritis	Nyu12	Nyu1200	69896004	4.1632E+14
312534011	[X]Seropositive rheumatoid arthritis, unspecified	Nyu1G	Nyu1G00	239791005	4.2651E+14
125937016	Acute polyarticular juvenile rheumatoid arthritis	N0431	N043100	75822003	125937016
371261000000111	Delivery of rehabilitation for rheumatoid arthritis	7P203	7P20300	2.3104E+14	3.7126E+14
1785241000006114	Except rheumatoid arthritis qual indicator: informed dissent	9hR1	9hR1.00	8.0598E+14	1785241000006114
1785231000006116	Except rheumatoid arthritis quality indicator: pt unsuitable	9hR0	9hR0.00	8.0596E+14	1785231000006116
311496011	Fibrosing alveolitis associated with rheumatoid arthritis	N04y0-2	N04y012	10713006	7.0951E+14

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309816012	Flare of rheumatoid arthritis	N040T	N040T00	201791009	309816012
309828016	Juvenile rheumatoid arthritis NOS	N043z	N043z00	410795001	7.5545E+14
2472447010	Juvenile rheumatoid arthritis	N0455	N045500	410795001	2472447010
755441000006119	Juvenile rheumatoid arthritis - Still's disease	N043	N043.00	410795001	7.5544E+14
297641010	Myopathy due to rheumatoid arthritis	F3964	F396400	193250002	297641010
309827014	Monarticular juvenile rheumatoid arthritis	N0433	N043300	201799006	309827014
123542016	Pauciarticular juvenile rheumatoid arthritis	N0432	N043200	74391003	123542016
297544012	Polyneuropathy in rheumatoid arthritis	F3712	F371200	193180002	297544012
1785221000006119	Exception reporting: rheumatoid arthritis quality indicators	9hR	9hR..00	8.0594E+14	1785221000006119
251794010	H/O: rheumatoid arthritis	14G1	14G1.00	161567008	251794010
309788014	Other rheumatoid arthritis of spine	N0401	N040100	9631008	1.6901E+13
451461014	Seropositive errosive rheumatoid arthritis	N047	N047.00	308143008	451461014
359292012	Seronegative rheumatoid arthritis	N040P	N040P00	239792003	359292012
149911000006117	Seropositive rheumatoid arthritis, unspecified	N04X	N04X.00	239791005	1.4991E+14
312529012	[X]Other specified arthritis	Nyu1B	Nyu1B00	3723001	4.1406E+14
250085017	FH: Rheumatoid arthritis	12I1	12I1.00	160413008	250085017
309789018	Rheumatoid arthritis of shoulder	N0402	N040200	201766009	309789018
2159844013	Rheumatoid arthritis particle agglutination test	43b9	43b9.00	1010071000000108	2582321000000111
162131000006111	Rheumatoid arthritis of PIP joint of finger	N0409	N040900	201773004	1.6213E+14
116082011	Rheumatoid arthritis	N040	N040.00	69896004	116082011
1875591000006110	Rheumatoid arthritis monitoring invitation first letter		EMISNQRH20	1875591000006106	1875591000006110
2196791000000119	Rheumatoid arthritis annual review	66HB0	66HB000	8.4726E+14	2196791000000119
2275501000000119	Rheumatoid arthritis monitoring invitation	9mM	9mM..00	8.8232E+14	2275501000000119
309794018	Rheumatoid arthritis of wrist	N0407	N040700	201771002	309794018
309800010	Rheumatoid arthritis of knee	N040D	N040D00	201777003	309800010
309805017	Rheumatoid arthritis of other tarsal joint	N040J	N040J00	429192004	1.6212E+14
309804018	Rheumatoid arthritis of talonavicular joint	N040H	N040H00	201781003	309804018

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309798015	Rheumatoid arthritis of hip	N040B	N040B00	201775006	309798015
309792019	Rheumatoid arthritis of elbow	N0405	N040500	201769002	309792019
162041000006117	Rheumatoid arthritis of DIP joint of finger	N040A	N040A00	201774005	1.6204E+14
162051000006115	Rheumatoid arthritis of distal radio-ulnar joint	N0406	N040600	201770001	1.6205E+14
309787016	Rheumatoid arthritis of cervical spine	N0400	N040000	201764007	309787016
309790010	Rheumatoid arthritis of sternoclavicular joint	N0403	N040300	201767000	309790010
309791014	Rheumatoid arthritis of acromioclavicular joint	N0404	N040400	201768005	309791014
426510015	Rheumatoid arthritis - multiple joint	N040S	N040S00	287006005	426510015
162101000006115	Rheumatoid arthritis of lesser MTP joint	N040L	N040L00	201784006	1.621E+14
162211000006111	Rheumatoid arthritis screen	68F1	68F1.00	1016811000000108	1.6221E+14
168761000006118	Rheumatoid arthritis of 1st MTP joint	N040K	N040K00	201783000	1.6876E+14
889731000006111	Rheumatoid arthritis NOS	N040-90	N040.90	69896004	8.8973E+14
162191000006110	Rheumatoid arthritis of tibio-fibular joint	N040E	N040E00	201778008	1.6219E+14
162081000006111	Rheumatoid arthritis of IP joint of toe	N040M	N040M00	201785007	1.6208E+14
162111000006117	Rheumatoid arthritis of MCP joint	N0408	N040800	201772009	1.6211E+14
162221000006115	Rheumatoid arthritis screening test	43c6	43c6.00	1016811000000108	2565831000000112
168751000006115	Rheumatoid arthritis and other inflammatory polyarthropathy	N04	N04..00	69896004	1.6875E+14
309802019	Rheumatoid arthritis of ankle	N040F	N040F00	201779000	309802019
309803012	Rheumatoid arthritis of subtalar joint	N040G	N040G00	201780002	309803012
909191000006119	[RFC] Rheumatoid arthritis	HNGP005		9.0919E+14	9.0919E+14
2275751000000111	Rheumatoid arthritis monitoring invitation third letter	9mM2	9mM2.00	8.8244E+14	2275751000000111
1875611000006116	Rheumatoid arthritis monitoring invitation third letter		EMISNQRH22	1875611000006100	1875611000006116
1875581000006112	Rheumatoid arthritis monitoring verbal invitation		EMISNQRH19	1875581000006108	1875581000006112
1875601000006119	Rheumatoid arthritis monitoring invitation second letter		EMISNQRH21	1875601000006103	1875601000006119
1880371000006117	Rheumatoid arthritis resolved		EMISNQRH23	1880371000006101	1880371000006117
1875571000006114	Rheumatoid arthritis monitoring telephone invitation		EMISNQRH18	1875571000006105	1875571000006114
2275831000000110	Rheumatoid arthritis monitoring telephone invitation	9mM4	9mM4.00	8.8248E+14	2275831000000110

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2275711000000112	Rheumatoid arthritis monitoring invitation second letter	9mM1	9mM1.00	8.8242E+14	2275711000000112
2275791000000115	Rheumatoid arthritis monitoring verbal invitation	9mM3	9mM3.00	8.8246E+14	2275791000000115
162141000006118	Rheumatoid arthritis of sacro-iliac joint	N040C	N040C00	201776007	1.6214E+14
424981000006114	[X]Rheumatoid arthritis+involvement/other organs or systems	Nyu10	Nyu1000	239793008	4.2498E+14
1875561000006119	Rheumatoid arthritis monitoring invitation		EMISNQRH17	1875561000006103	1875561000006119
2275671000000111	Rheumatoid arthritis monitoring invitation first letter	9mM0	9mM0.00	8.824E+14	2275671000000111
20587019	Acute arthritis	N06zA	N06zA00	11939005	20587019
359168010	Ankle arthritis NOS	N06z7-1	N06z711	239874001	4.8369E+14
310552015	Ankle joint ankylosis	N0857-1	N085711	202318002	310552015
16833013	Ankylosing spondylitis	N100	N100.00	9631008	16833013
484889013	Ankylosing vertebral hyperostosis	N117	N117.00	31487001	484889013
1488690018	FH: Ankylosing spondylitis	12I7	12I7.00	394990003	1488690018
2325761000000118	Bath Ankylosing Spondylitis Functional Index	388p0	388p000	9.0431E+14	2325761000000118
2326221000000110	Bath Ankylosing Spondylitis Metrology Index	38QL	38QL.00	9.0451E+14	2326221000000110
1743391000006117	Ankylosing vertebral hyperostosis	N117-z376		1743391000006101	1743391000006117
497061000006119	Atlanto-axial ankylosis	N1481	N148100	202813002	4.9706E+14
311288015	Atlanto-occipital instability	N1487	N148700	202819003	311288015
311281014	Atlanto-occipital ankylosis	N1480	N148000	202812007	311281014
497071000006114	Atlanto-axial instability	N1488	N148800	202820009	4.9707E+14
1898101000006110	Axial spondyloarthritis	EMISNQAX2		1898101000006106	1898101000006110
2360741000000115	Axial spondyloarthritis	N11F	N11F.00	723116002	3334645019
2692996011	BASDAI - Bath ankylosing spondylitis disease activity index	388p	388p.00	429712009	2692996011
1786505011	Caplan's syndrome	N04y0-1	N04y011	398640008	1786505011
311283012	Cervical spine ankylosis	N1482	N148200	202814008	311283012
311290019	Cervical spine instability	N1489	N148900	202821008	311290019
544821000006112	Cervico-thoracic instability	N148A	N148A00	202822001	5.4482E+14
544811000006116	Cervico-thoracic ankylosis	N1483	N148300	202815009	5.4481E+14

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59922014	Chronic arthritis	N06zB	N06zB00	35908007	59922014
359172014	Elbow arthritis NOS	N06z2-1	N06z211	439656005	6.3739E+14
95067016	Felty's syndrome	N041	N041.00	57160007	95067016
359167017	Foot arthritis NOS	N06z7-2	N06z712	448394006	7.6424E+14
310159018	Generalised arthritis	N065A	N065A00	202031002	310159018
815501000006110	Hand arthritis NOS	N06z4-1	N06z411	448589005	8.155E+14
359170018	Hip arthritis NOS	N06z5-1	N06z511	68449006	8.2375E+14
481879010	Hip joint ankylosis	N0855-1	N085511	24273009	481879010
310551010	Knee joint ankylosis	N0856-1	N085611	202317007	310551010
108781000006112	Temporomandibular joint ankylosis	J0460	J046000	50603008	1.0878E+14
310544018	Wrist joint ankylosis	N0853-1	N085311	202310009	310544018
310540010	Elbow joint ankylosis	N0852-1	N085211	202307002	310540010
312611014	[X]Other specified inflammatory spondylopathies	Nyu61	Nyu6100	202649003	4.156E+14
312615017	[X]Other specified spondylopathies	Nyu65	Nyu6500	4.1483E+14	4.1639E+14
311043013	Inflammatory spondylopathies in diseases EC	N10y0	N10y000	202652006	311043013
312610010	[X]Other infective spondylopathies	Nyu60	Nyu6000	721273004	4.1114E+14
311044019	Other inflammatory spondylopathies NOS	N10yz	N10yz00	202649003	2.8481E+13
311040011	Inflammatory spondylopathies	N10	N10..00	202649003	311040011
311042015	Other inflammatory spondylopathies	N10y	N10y.00	202649003	2.8471E+13
890821000006113	Inflam. spondylopathies NOS	N10z-99	N10z.99	6.8523E+14	8.9082E+14
310518018	Joint ankylosis of unspecified site	N0850	N085000	111227009	7.5789E+14
400222018	Joint ankylosis of the upper arm	N0852	N085200	267932007	400222018
310519014	Joint ankylosis of the shoulder region	N0851	N085100	202298000	310519014
310537010	Joint ankylosis of other specified site	N0858	N085800	111227009	7.5781E+14
400226015	Joint ankylosis of the ankle and foot	N0857	N085700	267936005	400226015
400223011	Joint ankylosis of the forearm	N0853	N085300	202310009	7.5783E+14
400225016	Joint ankylosis of the lower leg	N0856	N085600	267935009	400225016

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400224017	Joint ankylosis of the pelvic region and thigh	N0855	N085500	267934008	400224017
757841000006114	Joint ankylosis of the hand	N0854	N085400	56923001	7.5784E+14
309836013	Juvenile arthritis in ulcerative colitis	N0454	N045400	201807008	309836013
309833017	Juvenile arthritis in Crohn's disease	N0453	N045300	201805000	309833017
359310019	Juvenile arthritis in psoriasis	N0452	N045200	239803008	359310019
359312010	Juvenile ankylosing spondylitis	N0450	N045000	239805001	359312010
309824019	Juvenile rheumatoid arthropathy unspecified	N0430	N043000	410793008	7.5546E+14
2472449013	Juvenile seronegative polyarthritis	N0451	N045100	410797009	7.5547E+14
359169019	Knee arthritis NOS	N06z6-1	N06z611	371081002	7.537E+14
311287013	Lumbar spine ankylosis	N1486	N148600	202818006	311287013
311293017	Lumbar spine instability	N148C	N148C00	202824000	311293017
1235278014	Lumbosacral ankylosis	N1460	N146000	87823004	1235278014
299135012	Malleus ankylosis	F5520	F552000	194333008	299135012
718001000006112	Marie - Strumpell spondylitis	N100-1	N100.11	9631008	7.18E+14
255925015	O/E-hands-rheumatoid spindling	2G27	2G27.00	164456005	255925015
253940015	O/E - ankyl.spondyl.chest def.	2377	2377	162930007	253940015
299136013	Ossicle ankylosis (excluding malleus)	F5521	F552100	194334002	299136013
16911000006114	Other rheumatoid arthropathy + visceral/systemic involvement	N042	N042.00	239793008	1.6911E+13
502690012	Pauciarticular onset juvenile chronic arthritis	N0456	N045600	74391003	502690012
311398011	Periarthritis NOS	N21z1	N21z100	50921008	2.3385E+14
178460018	Periarthritis of wrist	N2142	N214200	111242007	178460018
1786700015	Periarthritis of shoulder	N2120	N212000	399114005	1786700015
221661000006114	Polyarthritis	N065z-1	N065z11	417373000	2.2166E+14
55628018	Psoriatic arthritis	M160-1	M160.11	33339001	55628018
55627011	Psoriatic arthropathy	M160	M160.00	33339001	55627011
308730016	Psoriatic arthropathy NOS	M160z	M160z00	33339001	1.9893E+14
198881000006116	Psoriasis spondylitica	M1600	M160000	200956002	1.9888E+14

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1756301000006119	Remitting seronegative symmetrical synovitis pitting oedema	N04y3	N04y300	445345005	1756301000006119
1746141000006118	Remitting seronegative symmetrical synovitis with pitting oedema		EMISNQRE122	1746141000006102	1746141000006118
39771016	Rheumatic carditis	N0420	N042000	23685000	39771016
162231000006117	Rheumatoid arthropathy + visceral/systemic involvement NOS	N042z	N042z00	239793008	1.6223E+14
309812014	Rheumatoid bursitis	N040Q	N040Q00	201788009	309812014
48361011	Rheumatoid carditis	G5yA	G5yA.00	28880005	48361011
1778239014	Rheumatoid lung disease	N0421	N042100	398726004	1778239014
162301000006117	Rheumatoid lung	H570	H570.00	398726004	1.623E+14
1786545019	Rheumatoid lung	N04y0	N04y000	398726004	1786545019
300221014	Rheumatoid myocarditis	G5y8	G5y8.00	195136004	300221014
311041010	Sacroiliitis NEC	N102	N102.00	55146009	1.5986E+14
311267012	Sacral ankylosis NOS	N1462	N146200	16065006	1.5957E+14
1221279016	Sacroiliac ankylosis	N1461	N146100	16065006	1221279016
311045018	Spondylitis NOS	N10z	N10z.00	84172003	1.2992E+14
17967011	Spinal enthesopathy	N101	N101.00	10317009	17967011
219271000000117	Sero negative arthritis	N04y1	N04y100	399112009	2.1927E+14
215551000000114	Sero negative polyarthritis	N04y1-1	N04y111	399112009	2.1555E+14
140551000006117	Shoulder arthritis NOS	N06z1-1	N06z111	443798008	1.4055E+14
559621000000119	Suspected inflammatory arthritis	1JG	1JG..00	431956005	2768203016
311285017	Thoracic spine ankylosis	N1484	N148400	202816005	311285017
311292010	Thoracic spine instability	N148B	N148B00	202823006	311292010
106931000006114	Thoraco-lumbar ankylosis	N1485	N148500	202817001	1.0693E+14
1464018	Unspecified monoarthritis	N066	N066.00	699462004	7.7391E+13
310170013	Unspecified monoarthritis of the hand	N0664	N066400	698475006	7.7441E+13
310172017	Unspecified monoarthritis of the lower leg	N0666	N066600	699462004	7.7451E+13
310169012	Unspecified monoarthritis of the forearm	N0663	N066300	699462004	7.7431E+13
310167014	Unspecified monoarthritis of the shoulder region	N0661	N066100	699462004	7.7471E+13

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310174016	Unspecified monoarthritis of other specified site	N0668	N066800	699462004	7.7411E+13
310171012	Unspecified monoarthritis of the pelvic region and thigh	N0665	N066500	699462004	7.7461E+13
310175015	Unspecified monoarthritis NOS	N066z	N066z00	699462004	7.7401E+13
310168016	Unspecified monoarthritis of the upper arm	N0662	N066200	699462004	7.7481E+13
310166017	Unspecified monoarthritis of unspecified site	N0660	N066000	699462004	7.7491E+13
310173010	Unspecified monoarthritis of the ankle and foot	N0667	N066700	699462004	7.7421E+13
405142017	Unspecified polyarthropathy or polyarthritis	N065	N065.00	36186002	7.8151E+13
400192018	Unspecified polyarthropathy or polyarthritis NOS	N065z	N065z00	36186002	7.8161E+13
359171019	Wrist arthritis NOS	N06z3-1	N06z311	1.771E+12	5.4521E+13

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Supplementary Table *S5.3*: Medcodeid used to define Juvenile-onset Systemic Lupus Erythematosus. Note that where adult lupus codes have been included, we also specified that the individual was under 18 at the point at which the code was observed.

Medcodeid	Term	Original Read Code	Cleansed Read Code	Snomed CT Concept Id	Snomed CT Description Id
92209015	Disseminated lupus erythematosus	N0000	N000000	55464009	92209015
92208011	Systemic lupus erythematosus	N000	N000.00	55464009	92208011
297542011	Polyneuropathy in disseminated lupus erythematosus	F3710	F371000	193178008	297542011
301721015	Lung disease with systemic lupus erythematosus	H57y4	H57y400	196138005	301721015
308701011	Discoid lupus erythematosus	M1541	M154100	200938002	308701011
309408013	Systemic lupus erythematosus NOS	N000z	N000z00	55464009	1.1424E+14
312579010	[X]Other forms of systemic lupus erythematosus	Nyu43	Nyu4300	55464009	4.1051E+14
512234017	Cerebral lupus	N0006	N000600	95644001	512234017
359447015	Subacute cutaneous lupus erythematosus	M1547	M154700	239891002	359447015
453243010	Systemic lupus erythematosus with pericarditis	N0004	N000400	309762007	453243010
309383019	[X]Other local lupus erythematosus	Myu78	Myu7800	7119001	4.1149E+14
114251000006118	Systemic lupus erythematosus with organ or sys involv	N0003	N000300	239887007	1.1425E+14
677561000006119	Nephrotic syndrome in systemic lupus erythematosus	K01x4	K01x400	68815009	6.7756E+14
660871000006118	Eyelid discoid lupus erythematosus	F4D33	F4D3300	79291003	6.6087E+14
25612013	Lupus erythematosus profundus	M1544	M154400	15084002	25612013
308706018	Lupus erythematosus tumidus	M1545	M154500	200941006	308706018
308697013	Lupus erythematosus	M154	M154.00	200936003	308697013
308709013	Lupus erythematosus NOS	M154z	M154z00	200936003	7.3214E+14
308700012	Lupus erythematosus chronicus	M1540	M154000	200937007	308700012
308704015	Lupus erythematosus migrans	M1542	M154200	200939005	308704015
308705019	Lupus erythematosus nodularis	M1543	M154300	200940007	308705019
308707010	Lupus erythematosus unguium mutilans	M1546	M154600	200942004	308707010
114310015	Lupus nephritis	K01x4-1	K01x411	68815009	114310015

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Supplementary Table S5.4: Medcodeid used to define Juvenile Dermatomyositis. Note that where adult codes have been included, we also specified that the individual was under 18 at the point at which the code was observed.

Medcodeid	Term	Original Read Code	Cleansed Read Code	Snomed CT Concept Id	Snomed CT Description Id
1776211019	Dermatomyositis	N003	N003.00	396230008	1776211019
312591018	[X]Dermatopolymyositis, unspecified	Nyu4E	Nyu4E00	419921000000105	939011000000110
614691000006113	Dermatopolymyositis, unspecified	N003X	N003X00	419921000000105	614691000006113
143230018	Poikilodermatomyositis	N003-1	N003.11	396230008	220521000006113
1784692011	Juvenile dermatomyositis	N0030	N003000	1212005	1784692011
312580013	[X]Other dermatomyositis	Nyu44	Nyu4400	396230008	408771000006110
301719013	Lung disease with polymyositis	H57y1	H57y100	196136009	301719013
52463013	Polymyositis	N004	N004.00	31384009	52463013

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Supplementary Table S5.5: ICD-10 codes for the following comorbidities were collated from published code lists and confirmed by the study team.

ICD10 Codes	Description
Alcohol misuse[1]	
F101	Alcohol abuse
F102	Alcohol dependence
Z714	Alcohol abuse counselling and surveillance
Autonomic neuropathy[2]	
G900	Idiopathic peripheral autonomic neuropathy
G9008	Other idiopathic peripheral autonomic neuropathy
G9009	Idiopathic peripheral autonomic neuropathy, unspecified
G99	
Benign essential tremor[2]	
G250	Essential tremor
Benign prostatic hyperplasia[3]	
N40	Benign prostatic hyperplasia without lower urinary tract symptoms
N40	Benign prostatic hyperplasia with lower urinary tract symptoms
Chronic cystitis[2]	
N300	Acute cystitis
N301	Interstitial cystitis (chronic)
N302	Other chronic cystitis
Chronic fatigue syndrome[2]	
G933	Chronic fatigue syndrome
Cirrhosis[4]	
K703	Alcoholic cirrhosis of liver
K743	
K744	Secondary biliary cirrhosis
K745	Biliary cirrhosis, unspecified
K746	Other and unspecified cirrhosis of liver
Congenital septal defect[2]	
Q219	Congenital malformation of cardiac septum, unspecified
Diabetic neuropathy[2]	
E084	
E094	

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E104	
E1140	Type 2 diabetes mellitus With neurological complications Controlled
E1141	Type 2 diabetes mellitus With neurological complications Uncontrolled
E134	
Fibromatosis[2]	
D4811	
M720	Palmar fascial fibromatosis [Dupuytren]
M722	Plantar fascial fibromatosis
M724	Pseudosarcomatous fibromatosis
M728	
Hypersplenism[2]	
D731	Hypersplenism
Idiopathic intracranial hypertension[2]	
G932	Benign intracranial hypertension
Infections[5]	
A00-B99	Infections (including parasitic diseases)
J00-J06	Acute upper respiratory infections
J09-J18	Influenza and pneumonia
J20-J22	Other acute lower respiratory infections
J32	Chronic sinusitis
J36	Peritonsillar abscess
J39.0-J39.1	Parapharyngeal abscess
J44.0	Chronic obstructive pulmonary disease with acute lower respiratory infection
J85-J86	Pyothorax and abscess of lung
L00-L08	Infections of the skin and subcutaneous tissue
M00-M03	Infectious arthropathies
N16.0	Renal tubule-interstitial disorders in infectious and parasitic diseases
N39.0	Urinary tract infection
N41.2	Abscess of prostate
N45	Orchitis and epididymitis
N73	Acute parametritis and pelvic cellulitis
N77.0	Ulceration of vulva in infectious and parasitic diseases
N77.1	Vaginitis, vulvitis and vulvovaginitis in infectious and parasitic diseases
J00-J06	Acute upper respiratory infections
J09-J18	Influenza and pneumonia
J20-J22	Other acute lower respiratory infections
J44.0	Chronic obstructive pulmonary disease with acute lower respiratory infection
J85-J86	Pyothorax and abscess of lung
L00-L08	Infections of the skin and subcutaneous tissue

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Multiple valve disorder[6]	
I080	Multiple valve diseases
I081-I087	
I088	Other multiple valve diseases
I089	Multiple valve disease, unspecified
Neuropathic bladder[2]	
N310	Uninhibited neuropathic bladder, not elsewhere classified
N311	Reflex neuropathic bladder, not elsewhere classified
N312	Flaccid neuropathic bladder, not elsewhere classified
Plasma cell malignancy[6]	
C900	Plasma cell myeloma
C901	Plasma cell leukaemia
C902-C903	
Primary malignancy_bowel	
C18-C21	
Rheumatic valve disorder[6]	
I05	Rheumatic mitral valve diseases
I06	Rheumatic aortic valve diseases
I07	Rheumatic aortic valve disease, unspecified
I08	Rheumatic tricuspid valve diseases
I09	
Right bundle branch block combinations[2]	
I451	Other and unspecified right bundle-branch block
I452-I453	
Scleroderma[7]	
M340	Localized Scleroderma
M341-M347	
M348	Scleredema Adultorum
M349	Scleredema Adultorum
Sickle cell disease[2]	
D570	Sickle-cell anaemia with crisis
D571	Sickle-cell anaemia without crisis
D572	Double heterozygous sickling disorders
D573	Sickle-cell trait
D574	
D578	Other sickle-cell disorders
Sjogren's syndrome[8]	
M3500	Sjogren syndrome, unspecified
M3501	Sjogren syndrome with keratoconjunctivitis
M3502	Sjogren syndrome with lung involvement

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M3503	Sjogren syndrome with myopathy
M3504	Sjogren syndrome with tubulo-interstitial nephropathy
M3505	Sjogren syndrome with inflammatory arthritis
M3506	Sjogren syndrome with peripheral nervous system involvement
M3507	Sjogren syndrome with central nervous system involvement
M3508	Sjogren syndrome with gastrointestinal involvement
M3509	Sjogren syndrome with other organ involvement
M350A	Sjogren syndrome with glomerular disease
M350B	Sjogren syndrome with vasculitis
M350C	Sjogren syndrome with dental involvement
Thyroid disease[2]	
E00-E07	
E070	Hypersecretion of calcitonin
E071	Dyshormogenetic goitre
E078	Other specified disorders of thyroid
E079	Disorder of thyroid, unspecified
Venous thromboembolic disease (Excl PE)[9]	
I278-I279	Venous thromboembolic disease (Excl PE)
I676	Venous thromboembolic disease (Excl PE)
I743	Venous thromboembolic disease (Excl PE)
I800	Venous thromboembolic disease (Excl PE)
I819-I823	Venous thromboembolic disease (Excl PE)
I828-I829	Venous thromboembolic disease (Excl PE)
K550	Venous thromboembolic disease (Excl PE)
O082	Venous thromboembolic disease (Excl PE)
O087	Venous thromboembolic disease (Excl PE)
O222	Venous thromboembolic disease (Excl PE)
O225	Venous thromboembolic disease (Excl PE)
O228-O229	Venous thromboembolic disease (Excl PE)
O870	Venous thromboembolic disease (Excl PE)
O872-O873	Venous thromboembolic disease (Excl PE)
O878-O881	Venous thromboembolic disease (Excl PE)
O883	Venous thromboembolic disease (Excl PE)
O888	Venous thromboembolic disease (Excl PE)

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Reference

1. **An online clinical codes repository to improve validity and reproducibility of medical database research** [https://clinicalcodes.rss.mhs.man.ac.uk/medcodes/article/201/codelist/res201-alcohol-misuse-icd10/]
2. **ICD code search** [https://gesund.bund.de/en/icd-code-suche]
3. **ICD-10-CM/PCS MS-DRG v37.2 Definitions Manual** [https://www.cms.gov/icd10m/version372-fullcode-cms/fullcode_cms/P0268.html]
4. Hayward KL, Johnson AL, McKillen BJ, Burke NT, Bansal V, Horsfall LU, Hartel G, Moser C, Powell EE, Valery PC: **ICD-10-AM codes for cirrhosis and related complications: key performance considerations for population and healthcare studies.** *BMJ Open Gastroenterol* 2020, **7**.
5. Yiu ZZN, Parisi R, Lunt M, Warren RB, Griffiths CEM, Langan SM, Ashcroft DM: **Risk of hospitalization and death due to infection in people with psoriasis: a population-based cohort study using the Clinical Practice Research Datalink.** *Br J Dermatol* 2021, **184:**78-86.
6. **Chapter IX. Diseases of the circulatory system (I00-I99).** [https://classbrowser.nhs.uk/ICD-10-5TH-Edition/vol1/block-i05-i09.htm]
7. De Almeida Chaves S, Derumeaux H, Do Minh P, Lapeyre-Mestre M, Moulis G, Pugnet G: **Assessment of the Accuracy of Using ICD-10 Codes to Identify Systemic Sclerosis.** *Clin Epidemiol* 2020, **12:**1355-1359.
8. **ICD-10-CM Code for Sjogren syndrome: M35.0. ICD-10 code M35.0 for Sjogren syndrome is a medical classification as listed by WHO under the range - Diseases of the musculoskeletal system and connective tissue.** [https://www.aapc.com/codes/icd-10-codes/M35.0?srsltid=AfmBOoohcxvHZAOXu7w9eO8Yyz34aDvUBCmDTB-UBbXYhzCVSwWPmI78]
9. Öhman L, Johansson M, Jansson J-H, Lind M, Johansson L: **Positive predictive value and misclassification of diagnosis of pulmonary embolism and deep vein thrombosis in Swedish patient registries.** *Clinical Epidemiology* 2018, **10:**1215-1221.

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